Statistical Methods & Machine Learning 1

Class #11

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Outline

- Problems with Model Errors – What do we do?
  - Zero Mean Assumption
  - Constant Variance Assumption
  - Normality Assumption
- Problems with Predictors
  - Collinearity
  - Variable Rescaling
- Lack of Fit

Announcements

- No quiz today!

Problems with Model Errors

What do we do?

Zero Mean Assumption

Zero Mean Error Violation

$E(\epsilon_i) \neq 0$

What is the problem?

- OLS estimators are not unbiased

  e.g., Cannot guarantee $E(\hat{\beta}_{OLS}) = \beta$

What to do?

- Include other predictors: e.g., polynomials or splines (Class #12), interactions
- Variable transformation (Class #12)

Constant Variance Assumption

Homoscedasticity (i.e., Constant Variance) Violation

$V(\epsilon_i) = \sigma^2 \text{ vs.} V(\epsilon_i) = \sigma_i^2$

What is the problem?

- Cannot guarantee that $\hat{V}(\hat{\beta})$ is unbiased of $\mathrm{V}(\hat{\beta})$
- Any problem with using $\hat{\beta}_{OLS}$?

What to do?

- Heteroscedastic consistent variance estimator (will calculate in lab)
- Weighted least squares estimation (will do in lab)
- Variable transformation (Class #12)

Weighted Least Squares Estimators - 1

OLS:

- $min_{\beta} \sum_{i=1} n(Y_i - \boldsymbol{X}_i\boldsymbol{\beta})^2$
- $\hat{\beta}_{OLS} = (\sum_{i=1}^{n} \boldsymbol{X}_i^T \boldsymbol{X}_i)^{-1} \sum_{i=1}^{n} \boldsymbol{X}_i^T Y_i = \boxed{(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}}$

WLS:

- $min_{\beta} \sum_{i=1} nw_i(Y_i - \boldsymbol{X}_i\boldsymbol{\beta})^2$ for a set of weights $w_i \geq 0$
- $\hat{\beta}_W = (\sum_{i=1}^{n} w_i\boldsymbol{X}_i^T \boldsymbol{X}_i)^{-1} \sum_{i=1}^{n} w_i\boldsymbol{X}_i^T Y_i = \boxed{(\boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{W}\boldsymbol{Y}}$
- $\boldsymbol{W} = diag(w_1, \cdots, w_n)$

OLS is a special case of WLS when $w_1 = \cdots = w_n = 1$ (i.e., $\boldsymbol{W} = \boldsymbol{I}_n$).

Weighted Least Squares Estimators - 2

What does WLS do?

- We are implicitly regressing $\sqrt{w_i}Y_i$ on $\sqrt{w_i}X_i$
- Residuals become $\sqrt{w_i}\hat{\epsilon}_i$

How do we find w_i?

- **<u>Not easy!</u>**
- $V(\epsilon) \propto X_i \rightarrow w_i = X_i^{-1}$
- Use the absolute residuals
- Use the squared residuals

Observations with low variability get high weight and those with high variability get low weight.

Weighted Least Squares Estimators - 3

What does WLS do?

- We are implicitly regressing $\sqrt{w_i}Y_i$ on $\sqrt{w_i}X_i$
- Residuals become $\sqrt{w_i}\hat{\epsilon}_i$

How do we find w_i?

- <u>**Not easy!**</u>
- $V(\epsilon) \propto X_i \;\rightarrow w_i = X_i^{-1}$
- Other common approaches
  - Use the absolute residuals
  - Use the squared residuals

Might choose this option after seeing positive relationship in plot of $|\hat{\varepsilon}_i|$ against x_i.

Normality Assumption

Non-normal Distribution of Errors

- What is the main issue if we reject the hypothesis of normally distributed errors?
- Problem: Cannot use the t-tests for $H_0: \beta_j = 0$
- Solution: Bootstrap standard error (more in Class #14) or Robust Regression (see Faraway text)

Problems with Predictors

Collinearity

Collinearity (i.e., Lack of identifiability)

- When some predictors are linear combinations of other predictors.
  - Called “exact collinearity”
- We’ve seen this before as a “lack of identifiability”.

Collinearity (i.e., Lack of identifiability)

Using the `Wage` dataset from the `ISLR2` package we'll fit a simple linear regression model with `wage` as a function of `jobclass` (specified as 2 dummy variables).

We'll create our dummy variables manually in this case. Specifically:

- indus: 1 = industrial; 0 = non-industrial
- info: 1 = information; 0 = non-information

```
lm(wage ~ indus + info, Wage)
```

Collinearity (i.e., Lack of identifiability)

```
Call:
lm(formula = wage ~ indus + info, data = wage)

Residuals:
     Min       1Q   Median       3Q      Max
-100.507  -25.362   -6.117   15.697  197.750

Coefficients: (1 not defined because of singularities)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  120.593      1.070  112.69   <2e-16 ***
indus        -17.272      1.492  -11.58   <2e-16 ***
info              NA         NA      NA       NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.83 on 2998 degrees of freedom
Multiple R-squared:  0.04281,    Adjusted R-squared:  0.04249
F-statistic: 134.1 on 1 and 2998 DF,  p-value: < 2.2e-16
```

Collinearity (i.e., Lack of identifiability)

```
> head(model.matrix(mod_j1))
       (Intercept) indus info
231655           1     1    0
86582            1     0    1
161300           1     1    0
155159           1     0    1
11443            1     0    1
376662           1     0    1
```

Both of our predictors are dummy/indicator variables derived from the `jobclass` variable.

Does anything stand out?

They're simply the inverse of each other. No additional information is provided by one that's not already provided by the other. We're trying to use 3 parameters to model only 2 groups.

Collinearity (i.e., Lack of identifiability)

- For a model $Y = X\beta + \epsilon$ with $\epsilon \sim N_n(\epsilon, \sigma^2 I_n)$ to be estimable X needs to be of full column rank (i.e., no repeating information)
- $X_i = (1, X_{i1}, \dots, X_{ip})$, for a subject i on p variables, is referred as collinear if X_{ij} can be expressed as linear combination of X's
- Effect of collinearity

$V(\hat{\beta}_j) = \sigma^2 \frac{1}{1-R_j^2} \frac{1}{\sum_{i=1}^n (X_{ij}-\bar{X}_j)^2}$, where

  - $\bar{X}_j = n^{-1} \sum_{i=1}^n X_{ij}$
  - R_j^2: R^2 from a model regressing X_j on other X's and the intercept

- No collinearity $\Leftrightarrow R_j^2 = 0 \Leftrightarrow V(\hat{\beta}_j) = \frac{\sigma^2}{\sum_{i=1}^n (X_{ij}-\bar{X}_j)^2} = \frac{\sigma^2}{S_{XX}}$
- Not no collinearity $\Leftrightarrow 0 < R_j^2 \leq 1 \frac{\sigma^2}{S_{XX}} < V(\hat{\beta}_j) \leq \infty$

How do we handle collinearity?

- Exact collinearity collinearity can be dealt with by removing one or more of the problematic variables → e.g., the reference group dummy variable.

How do we handle collinearity?

- Exact collinearity can be dealt with by removing one or more of the problematic variables → e.g., the reference group dummy variable.
- What about when the collinearity is not exact?
  - This is more difficult and requires careful assessment of how severe the collinearity (or multicollinearity) is.

Why might non-exact collinearity be problematic?

Collinearity Indicators

- To understand why collinearity (even when non-exact) can be problematic let’s look at one of the indicators used to assess it.
  - Variance Inflation Factor

Collinearity Indicator - Variance Inflation Factor (VIF)

For a predictor $X_j, V(\hat{\beta}_j) = \sigma^2 \frac{1}{1-R_j^2} \frac{1}{(X_{ij}-\bar{X}_j)^2}$, where

R_j^2: R^2 from a model regressing X_j on other X's and the intercept

Variance Inflation Factor: $VIF_j = \frac{1}{1-R_j^2}$

When X_j is orthogonal to other predictors

$\Leftrightarrow R_j^2=0 \Leftrightarrow VIF_j=1$

When X_j is highly correlated with other predictors

$\Leftrightarrow R_j^2 \to 1 \Leftrightarrow VIF_j \to \infty$

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When X_j is highly correlated with other predictors

$\Leftrightarrow R_j^2 \rightarrow 1 \Leftrightarrow VIF_j \rightarrow \infty$

What would you expect R_j^2 to look like when there is substantial collinearity?

Collinearity Indicator - Variance Inflation Factor (VIF)

For a predictor $X_j, V\left(\hat{\beta}_j\right) = \sigma^2 \frac{1}{1-R_j^2} \frac{1}{(X_{ij}-\bar{X}_j)^2}$, where

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When X_j is highly correlated with other predictors

$\Leftrightarrow R_j^2 \rightarrow 1 \Leftrightarrow VIF_j \rightarrow \infty$

More collinearity → Larger R_j^2
Less collinearity → Smaller R_j^2

What does this mean for the variance of our predictors?

Collinearity Indicator - Variance Inflation Factor (VIF)

For a predictor $X_j, V\left(\hat{\beta}_j\right) = \sigma^2 \frac{1}{1-R_j^2} \frac{1}{(X_{ij}-\bar{X}_j)^2}$, where

R_j^2: R^2 from a model regressing X_j on other X's and the intercept

Variance Inflation Factor: $VIF_j = \frac{1}{\quad}$

More collinearity → Larger R_j^2 → Smaller denominator → Larger VIF
Less collinearity → Smaller R_j^2 → Larger denominator → Smaller VIF

$\Leftrightarrow R_j^2$=0 $\Leftrightarrow VIF_j$=1

When X_j is highly correlated with other predictors

$\Leftrightarrow R_j^2 \to 1 \Leftrightarrow VIF_j \to \infty$

Collinearity Indicator - Variance Inflation Factor (VIF)

For a predictor $X_j, V(\hat{\beta}_j) = \sigma^2 \frac{1}{1-R_j^2} \frac{1}{(X_{ij}-\bar{X}_j)^2}$, where

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When X_j is orthogonal to other predictors

$\Leftrightarrow R_j^2$=0 $\Leftrightarrow VIF_j$=1

When X_j is highly correlated with other predictors

$\Leftrightarrow R_j^2 \rightarrow 1 \Leftrightarrow VIF_j \rightarrow \infty$

So, what would we expect to see with high collinearity/large VIF?

Collinearity example

Let's look at an example using the `seatpos` data from the `faraway` package

- Data on 38 drivers regarding where they prefer their seat position along with age and measures of physical size (weight, length of extremities, etc.)
- We'll fit a model with `hipcenter` (distance in mm from hips to fixed location in car) as a function of the other 8 variables in the dataset.
  - `lmod <- lm(hipcenter ~ ., seatpos)`

Collinearity example

- Before we look at the model and VIF let's look at the correlation matrix for the predictors.
- Anything stand out?

```
> round(cor(seatpos[,-9]),2)
          Age Weight HtShoes    Ht Seated  Arm Thigh   Leg
Age      1.00   0.08   -0.08 -0.09  -0.17 0.36  0.09 -0.04
Weight   0.08   1.00    0.83  0.83   0.78 0.70  0.57  0.78
HtShoes -0.08   0.83    1.00  1.00   0.93 0.75  0.72  0.91
Ht      -0.09   0.83    1.00  1.00   0.93 0.75  0.73  0.91
Seated  -0.17   0.78    0.93  0.93   1.00 0.63  0.61  0.81
Arm      0.36   0.70    0.75  0.75   0.63 1.00  0.67  0.75
Thigh    0.09   0.57    0.72  0.73   0.61 0.67  1.00  0.65
Leg     -0.04   0.78    0.91  0.91   0.81 0.75  0.65  1.00
> 
```

Collinearity example

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- Anything stand out?

```
> round(cor(seatpos[,-9]),2)
          Age Weight HtShoes    Ht Seated  Arm Thigh   Leg
Age      1.00   0.08   -0.08 -0.09  -0.17 0.36  0.09 -0.04
Weight   0.08   1.00    0.83  0.83   0.78 0.70  0.57  0.78
HtShoes -0.08   0.83    1.00  1.00   0.93 0.75  0.72  0.91
Ht      -0.09   0.83    1.00  1.00   0.93 0.75  0.73  0.91
Seated  -0.17   0.78    0.93  0.93   1.00 0.63  0.61  0.81
Arm      0.36   0.70    0.75  0.75   0.63 1.00  0.67  0.75
Thigh    0.09   0.57    0.72  0.73   0.61 0.67  1.00  0.65
Leg     -0.04   0.78    0.91  0.91   0.81 0.75  0.65  1.00
>
```

There are some pretty large pairwise correlations! Ht:HtShoes, Seated:Ht, Seated:HtShoes, Ht:Weight, among others.

Collinearity example

- Let's look at the VIF for each predictor using the `vif()` function from the `faraway` package.
- Unsurprisingly Ht and HtShoes have the largest VIF suggesting a large amount of variance inflation.
- Specifically, taking $\sqrt{VIF}$ will let us approximate how much larger (i.e., inflated) the standard error of a given predictor is with this collinearity than if no collinearity was present in the model.

```
> vif(lmod)
       Age     Weight    HtShoes         Ht     Seated        Arm      Thigh        Leg
  1.997931   3.647030 307.429378 333.137832   8.951054   4.496368   2.762886   6.694291
```

Collinearity example

- Let' … edictor us … om the far…
- Uns … have the … ge am…
- Spe … t us approximate how much larger (i.e., inflated) the standard error of a given predictor is with this collinearity than if no collinearity was present in the model.

$\sqrt{307.43} = 17.5$ → the standard error for `HtShoes` in the model is 17.5 times larger than it would have been without collinearity.

$\sqrt{333.14} = 18.3$ → the standard error for `Ht` in the model is 18.3 times larger than it would have been without collinearity.

```
> vif(lmod)
      Age     Weight     HtShoes          Ht    Seated       Arm     Thigh       Leg
 1.997931   3.647030 307.429378  333.137832  8.951054  4.496368  2.762886  6.694291
```

Collinearity example

$\sqrt{307.43} = 17.5 \rightarrow$ the
standard
HtShoe
17.5 tim
would have been without
collinearity.

$\sqrt{333.14} = 18.3 \rightarrow$ the
larger than it would have
been without collinearity.

While we can't use this as a correction, it does give us a good sense of how severe the effect is.

```
> vif(lmod)
     Age   Weight    HtShoes         Ht    Seated      Arm    Thigh      Leg
1.997931 3.647030 307.429378 333.137832  8.951054 4.496368 2.762886 6.694291
```

Other indicators of collinearity

Collinearity Indicators

1. Variance Inflation Factor: Measure and assess existence of collinearity
2. Correlation among predictors: Assess existence of collinearity
3. Eigenvalues: Measure and understand collinearity
4. Condition Number/Index: Measure and understand collinearity

Collinearity Indicators

1. Variance Inflation Factor: Measure and assess existence of collinearity
2. Correlation among predictors: Assess existence of collinearity
3. Eigenvalues: Measure and understand collinearity
4. Condition Number/Index: Measure and understand collinearity

Collinearity Indicator 2 - Eigenvalues

OLS Estimators: $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y} \quad V(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$

Spectral decomposition: $\boldsymbol{X}^T\boldsymbol{X} = \boldsymbol{V}\boldsymbol{D}^2\boldsymbol{V}^T$, where

$\boldsymbol{V}$ is a $p \times p$ matrix such that $\boldsymbol{V}\boldsymbol{V}^T = \boldsymbol{I}_p$,

$\boldsymbol{D}^2$ is a $p \times p$ diagonal matrix such that $\boldsymbol{D}^2 = diag(d_1^2, \cdots, d_p^2)$,

$d_1^2, \cdots, d_p^2$ values are the eigenvalues of $\boldsymbol{X}^T\boldsymbol{X}$,

the columns of $\boldsymbol{V}$ contain the associated eigenvectors.

Also, $(\boldsymbol{X}^T\boldsymbol{X})^{-1} = \boldsymbol{V}\boldsymbol{D}^{-2}\boldsymbol{V}^T$, where

$\boldsymbol{D}^{-2} = diag(d_1^{-2}, \cdots, d_p^{-2})$

$d_1^{-2}, \cdots, d_p^{-2}$ are the eigenvalues of $(\boldsymbol{X}^T\boldsymbol{X})^{-1}$

Eigenvalues provide information about the behavior of OLS estimates as

$V(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1} = \sigma^2\boldsymbol{V}\boldsymbol{D}^{-2}\boldsymbol{V}^T$

Therefore, $d_j^2 \to 0 \quad \Leftrightarrow \quad d_j^{-2} \to \infty \quad \Leftrightarrow \quad V(\hat{\beta}_j) \to \infty$

Collinearity Indicator 2 - Eigenvalues

OLS Estimators: $\hat{\beta} = (X^T X)^{-1} X^T Y \quad V(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$

Spectral decomposition: $X^T X = V D^2 V^T$, where

V is a $p \times p$ matrix such that $V V^T = I_p$,

D^2 is a $p \times p$ diagonal matrix such that $D^2 = diag(d_1^2, \cdots, d_p^2)$

- Eigenvalue of 0 indicates exact collinearity.
- Several small eigenvalues suggests multicollinearity.
- Smaller eigenvalue → more problematic collinearity.

Also, $(X^T X)^{-1} = V D^{-2} V^T$, where

$D^{-2} = diag(d_1^{-2}, \cdots, d_p^{-2})$

$d_1^{-2}, \cdots, d_p^{-2}$ are the eigenvalues of $(X^T X)^{-1}$

Eigenvalues provide information about the behavior of OLS estimates as

$V(\hat{\beta}) = \sigma^2 (X^T X)^{-1} = \sigma^2 V D^{-2} V^T$

Therefore, $d_j^2 \to 0 \quad \Leftrightarrow \quad d_j^{-2} \to \infty \quad \Leftrightarrow \quad V(\hat{\beta}_j) \to \infty$

Collinearity Indicator 4 – Condition Number/Kappa

Overall measure of the eigenvalues of $X^T X$

Condition number: $\mathcal{K} = \sqrt{\frac{max_{(1 \le j \le p)} d_j}{min_{(1 \le j \le p)} d_j}}$

Condition number ($\mathcal{K}$)/indices (n_j) provide a means for identifying whether there are concerning eigenvalues.

Large $\mathcal{K}$: At least one of the eigenvalues is small relative to the rest

Condition index: $n_j = \sqrt{\frac{max_{(1 \le j \le p)} d_j}{d_j}}$ for $j = 1, \ldots, p$

Largest $n_j = \mathcal{K}$

Condition number ($\mathcal{K}$)≥30 is considered large.
Still worth examining other condition indices (n_j), as there may be more than one linear combination contributing to the collinearity.

Practical Guidelines for Collinearity

Diagnostics

1. Find $\mathcal{K} \geq 30$ and examine other n_j
2. Examine VIF
3. Examine correlation – Remember this is only looking at pairwise correlation

What to do?

- Remove predictors(s) with large collinearity → not always ideal/feasible
- Ridge regression – see SURV 616/SURVMETH 686

Collinearity – Summing Up

- What does the formula below actually tell us?
- $condition\ number = \sqrt{\max(eigenvalue)\ /eigenvalue_i}$
  - The higher the condition number the greater the difference between your largest eigenvalue and the eigenvalue in the denominator
- Kappa is simply your largest condition number or
  - kappa= $\sqrt{\max(eigenvalue)\ /\min(eigenvalue)}$
  - We should investigate if kappa > 30
  - Does kappa >30 mean there is definitely collinearity?

Collinearity – Summing Up

- What d
- *condit*
  - The
    eig
- Kappa
  - kap
  - We
  - Do

No! Large condition numbers and/or a kappa > 30 only tells us that at least one of our eigenvalues is much smaller than our largest eigenvalue

This could be due to collinearity – denominator of condition number formula is close to 0 leading to large condition number/kappa

Could also be driven by very large maximum eigenvalue – numerator of condition number is large leading to large condition number/kappa

Collinearity – Summing Up

- Should look at what potential collinearity is doing to our variance
  - Do we need to drop a variable?
  - If so, which one(s)?

$$VIF(\hat{\beta}_j) = \frac{1}{1 - R^2_{X_j|X_{-j}}}$$

- Rule of thumb in the social sciences is predictors with VIF > 5 are likely problematic
- Can also roughly interpret $\sqrt{VIF}$ as how much larger we would expect our standard error for that variable to be compared to if there were no collinearity
  - E.g., $VIF = 307.4 \ \rightarrow \ \sqrt{307.4} = 17.5$ would expect standard error to be 17.5 times larger than if no collinearity → while useful can't be applied as a correction because didn't actually observe orthogonal data

Variable Rescaling

Rescaling Predictors (and Outcomes)

Why rescaling?

- The range is really large or really small

→Estimated coefficients may be too small or too large

- Different measurement systems

→Interpretation of estimated coefficients may not be straight-forward

- Different relationship between Y and a numeric X depends on the range of X

→Recode X as a categorical variable

- Rescaling does not change hypothesis testing, but does change the magnitudes of model parameters
- NOT TRANSFORMATION

Rescaling the outcome – Example

- Fit model of median value of owner-occupied home in $1000 (`medv` in `Boston` dataset) predicted by `tax` (full property tax rate per $10,000)

```
mod1 <- lm(medv ~ tax, Boston)
```

- Put `medv` back on original scale and fit model

```
mod2 <- lm((medv*1000) ~ tax, Boston)
```

Rescaling your outcome

Rescaling your outcome doesn't impact the relationship, only the units on which it's being assessed.

```
Call:
lm(formula = medv ~ tax, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-14.091  -5.173  -2.085   3.158  34.058 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 32.970654   0.948296   34.77   <2e-16 ***
tax         -0.025568   0.002147  -11.91   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.133 on 504 degrees of freedom
Multiple R-squared:  0.2195,	Adjusted R-squared:  0.218 
F-statistic: 141.8 on 1 and 504 DF,  p-value: < 2.2e-16
```

```
Call:
lm(formula = (medv * 1000) ~ tax, data = Boston)

Residuals:
   Min     1Q Median     3Q    Max 
-14091  -5173  -2085   3158  34058 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 32970.654    948.296   34.77   <2e-16 ***
tax           -25.568      2.147  -11.91   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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Rescaling your predictor

- Fit model of median value of owner-occupied home in $1000 (`medv` in `Boston` dataset) predicted by `tax` (full property tax rate per $10,000)

```
mod1 <- lm(medv ~ tax, Boston)
```

- Put `tax` back on original scale and fit model

```
Boston$tax2 <- Boston$tax*10000

mod3 <- lm(medv ~ tax2, Boston)
```

Rescaling your predictor

Rescaling your predictor doesn't impact the relationship, only the units on which it's being assessed.

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Residuals:
    Min      1Q  Median      3Q     Max 
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```
Call:
lm(formula = medv ~ tax2, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-14.091  -5.173  -2.085   3.158  34.058 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.297e+01  9.483e-01   34.77   <2e-16 ***
tax2        -2.557e-06  2.147e-07  -11.91   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

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F-statistic: 141.8 on 1 and 504 DF,  p-value: < 2.2e-16
```

Lack of Fit

Lack of fit

- We've talked about how we might compare fit between models, but what about fit of a given model?
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- We've talked about how we might compare fit between models, but what about fit of a given model?
  - Ideally we would compare $\hat{\sigma}^2$ to σ^2 -- so why don't we?
  - Don't generally know σ^2
- So, what can we do?

Lack of Fit

- Need to compare $\hat{\sigma}^2$ to some model-free estimator of σ^2
- When we have repeated values can use this to our advantage
  - Must be independent (i.e., can't be repeated measures of same individuals)!

n values for a predictor variable, and

m_i observations of the outcome variable for the i^{th} value of the predictor variable

$Y_{ij} = \beta_0 + \beta_1 X_{1i} + \epsilon_{ij},$

Where $i = 1, \ldots, n; j = 1, \ldots, m_i$ and $\epsilon_{ij} \sim N(0, \sigma^2)$

$\rightarrow \bar{Y}_i = \frac{\sum_{i=1}^{m_i} Y_{ij}}{m_i} \sim N(\beta_0 + \beta_1 X_{1i}, \frac{\sigma^2}{m_i})$

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Let y_{ij} represent the i^{th} observation in a group of true replicates j

The pure error (model-free estimate of σ^2) is given by $\frac{SS_{pe}}{df_{pe}}$ and

$$SS_{pe} = \sum_i \sum_j (y_{ij} - \bar{y}_j)^2$$

Where $\bar{y}_j$ is the mean within replicate group j.

$df_{pe} = \#replicates_j - 1 = n - \#\ groups$

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So how do go about getting this model-free estimate of σ^2 AKA pure error?

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Lack of Fit

Estimating "pure error"

- Fit a model where a parameter is estimated for each group of observations for a given value of x.
  - We can do this in R by wrapping a continuous predictor in the `factor()` function when we fit the model.
- The $\hat{\sigma}^2$ from this model will be our "pure error".
- This model in itself tells us nothing interesting about the predictors relationship with the response, it simply fits a mean to each group of replicates.
  - Values of x without replicates are estimated exactly and don't contribute to $\hat{\sigma}^2$.
- We can then compare this model with the regression model of interest using an F test.

Lack of Fit Testing - 1

$$
\begin{aligned}
RSS &= \sum_{i=1}^{n}\sum_{j=1}^{m_i}(Y_{ij}-\hat{Y}_i)^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m_i}((Y_{ij}-\bar{Y}_i)+(\bar{Y}_i-\hat{Y}_i))^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m_i}(Y_{ij}-\bar{Y}_i)^2+\sum_{i=1}^{n}\sum_{j=1}^{m_i}(\bar{Y}_i-\hat{Y}_i)^2 \\
&\quad +2\sum_{i=1}^{n}\sum_{j=1}^{m_i}(Y_{ij}-\bar{Y}_i)(\bar{Y}_i-\hat{Y}_i) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m_i}(Y_{ij}-\bar{Y}_i)^2+\sum_{i=1}^{n}m_i(\bar{Y}_i-\hat{Y}_i)^2 \\
&= \boxed{RSS_{PureError}+RSS_{LackofFit}}
\end{aligned}
$$

Lack of Fit Testing - 2

When using all m_i observations,

$$RSS_{PureError} = \sum_{i=1}^{n} \sum_{j=1}^{m_i} (Y_{ij} - \bar{Y}_i)^2$$

$$\text{with } df_{PureError} = df_{TotalError} - df_{LackofFit} = \sum_{i=1}^{n} (m_i - 1)$$

$$\rightarrow RSS_{PureError} \sim \sigma^2 \chi^2_{df_{PureError}}$$

$$RSS_{LackofFit} = \sum_{i=1}^{n} m_i (\bar{Y}_i - \hat{Y}_i)^2$$

$$\text{with } df_{LackofFit} = df_{TotalError} - df_{PureError} = n - 2$$

$$\rightarrow RSS_{LackofFit} \sim \sigma^2 \chi^2_{df_{LackofFit}}$$

As $RSS_{PureError}$ and $RSS_{LackofFit}$ are independent,

F-test of lack of fit:

$$F = \frac{MSE_{LackofFit}}{MSE_{PureError}} = \frac{RSS_{LackofFit}/df_{LackofFit}}{RSS_{PureError}/df_{PureError}} \sim F^{df_{LackofFit}}_{df_{PureError}}$$

Lab Session

What should we learn today?

- Problems with predictors
  - Checking collinearity: 1) Eigenvalues; 2) Variance Inflation Factor (VIF); 3) Condition number ($\mathcal{K}$) and index (n_j)
  - Variable rescaling
- Problems with model errors
  - Zero mean assumption (Solutions Class #12)
  - Constant variance (i.e., homoscedasticity) assumption

    →Solution: 1) Heteroscedastic consistent variance estimator; 2) Weighted linear regression; 3) Variable transformation (Class #12)
  - Normality assumption
  - Lack of fit